USE CASE DIAGRAM

Project Title	AI Academic Mentor
Team	CSE4104-7D-T06
Section	7D
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Team Information

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USE CASE DIAGRAM



Figure: Use Case Diagram — AI Academic Mentor System

USE CASE DIAGRAM — DESCRIPTION & ANALYSIS

1. Overview

The Use Case Diagram for the AI Academic Mentor System provides a comprehensive visual representation of all functional interactions between the system's actors and its core features. The diagram is designed in compliance with standard UML notation and captures the complete behavioral scope of the system from both user and administrative perspectives. It clearly delineates how the system's three primary actors — Student, Admin, and AI Services (Gemini) — interact with the various use cases defined within the system boundary.

2. System Boundary

The system boundary, labeled "AI Academic Mentor System," encloses all the defined use cases and establishes the scope of the application under development. Every feature or function that the system is responsible for is represented within this boundary, while external actors remain outside it. This boundary-based modeling ensures a clear separation between internal system functionality and external entities that communicate with the system.

3. Actors

3.1 Student (Primary Actor)

The Student actor represents the primary end-user of the system. Students interact with the largest set of use cases, encompassing both core AI-driven learning tools and supporting account management features. The Student actor initiates most of the system’s functional operations, including account registration, AI chatbot interaction, quiz generation, note summarization, and study plan creation.

3.2 Admin (Secondary Actor)

The Admin actor represents the system administrator who oversees platform operations. The Admin has access to backend management features such as monitoring system usage, viewing all registered users, posting announcements, and managing user accounts. Admin functions are essential for maintaining the integrity, security, and smooth operation of the platform.

3.3 AI Services — Gemini (External Actor)

The AI Services actor, powered by Google’s Gemini API, is an external system that the AI Academic Mentor integrates with to provide intelligent responses and content generation. This actor is connected via include/extend relationships to core AI features such as chatbot interaction, quiz generation, and note summarization, highlighting the system’s dependency on an external AI backend for its intelligent capabilities.

4. Use Cases — Student Module

4.1 Register Account & Login Account

These foundational use cases allow new users to create an account on the platform and subsequently authenticate themselves to access the system’s features. Registration captures user credentials and basic information, while login validates identity and establishes a session. These use cases form the entry point for all student interactions with the system.

4.2 Interact with AI Chatbot

This is the central use case of the Student module. Students can engage in real-time academic conversations with an AI-powered chatbot that leverages Gemini API responses. The chatbot is designed to answer curriculum-specific questions, provide explanations, and offer academic support. This use case extends to include conversation history viewing, enabling students to revisit past interactions and track their learning journey.

4.3 Generate AI Quiz

Students can request the generation of personalized quizzes based on specific topics or subjects. The AI generates contextually relevant questions, and students can attempt the quiz within the platform. An included sub-use case, "Attempt Quiz & View Score," allows students to submit their answers and receive immediate performance feedback, contributing to the progress dashboard data.

4.4 AI Note Summarizer

This use case enables students to submit lengthy notes or academic content for AI-driven summarization. The system processes and condenses the material into concise, digestible summaries. The "View Saved Summaries" sub-use case allows students to access previously generated summaries at any time, promoting continuous and efficient study.

4.5 Create Personalized Study Plan

Based on user input and learning goals, the system generates a tailored study plan for the student. The "Modify Study Plan" sub-use case gives students the flexibility to adjust their study schedule and goals dynamically, ensuring the plan remains aligned with their academic needs and progress.

4.6 Update Profile & View Progress Dashboard

Students can manage their personal profile information through the Update Profile use case. The View Progress Dashboard use case aggregates data from quiz scores, chatbot activity, and study plans to provide a holistic view of the student's academic progress over time, fostering self-awareness and goal-oriented learning.

5. Use Cases — Admin Module

5.1 Admin Login

This use case provides a secure, role-specific authentication entry point for administrators. Admin credentials are validated separately from student credentials to maintain access control integrity and protect sensitive system management functions.

5.2 Monitor System Usage

Admins can track platform-wide activity, including the number of active users, chatbot interaction frequency, quiz attempts, and other usage metrics. An included sub-use case, "View User Activity," provides granular insight into individual user behavior, which is essential for performance optimization and identifying misuse.

5.3 Post Announcements

This use case enables administrators to publish system-wide announcements or notifications to students. These announcements may include maintenance alerts, new feature rollouts, academic deadlines, or institutional updates. It corresponds to the "View Notifications" use case on the student side.

5.4 View All Users & Manage User Accounts

Admins can retrieve a complete list of registered users on the platform. The "Manage User Accounts" sub-use case further allows admins to perform actions such as activating, deactivating, or removing user accounts, ensuring proper governance of the user base.